

Notes: Eren & Malamud (JFE, 2022)

Dominant Currency Debt

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1 Big picture

- **Question.** Why do large global firms issue so much international debt in dollars rather than in other deep, liquid currencies such as the euro, yen, or Swiss franc?
- **Debt view.** The paper proposes a borrower-driven explanation: firms choose dollar debt because dollar repayment burdens become easier in long-run downturns.
- **Dominant currency.** The dominant debt currency is the one that maximizes the issuer’s tradeoff between tax benefits and default risk, after issuance costs.
- **Key result.** Conditional on issuance costs, firms prefer to issue in the currency with the highest covariance with firm value or the stock market over debt-maturity horizons.
- **Dollar paradox.** The dollar may be a safe haven at short horizons but a borrower hedge at longer horizons.
- **Inflation channel.** If relative PPP matters, inflation risk premia and monetary policy expectations become central to debt-currency choice.
- **Evidence.** QRP, IRP, historical dollar-stock betas, maturity regressions, yield-curve regressions, and COVID issuance patterns all support the debt view.

2 Data and measurement

The paper uses BIS international debt securities data for amounts outstanding and gross issuance by currency, Kremens–Martin quanto-implied risk premia, Hordahl–Tristani inflation risk premia, FRED trade-weighted dollar index returns, S&P 500 and MSCI AC World returns, and granular bond-issuance data. The main outcome is the share or indicator of dollar-denominated debt.

Interpretation. The data are chosen to connect the theory’s pricing object—expected currency-stock covariance at debt horizons—to the quantity outcome: actual currency choice in debt issuance.

3 Theory and empirical specifications

3.1 Capital structure and currency choice

A large global firm issues defaultable nominal debt in multiple currencies. Debt creates tax benefits but increases expected default costs. Currency denomination affects the dollar value of future repayment obligations.

$$B_{t+1}(B_t) = ((1 - \tau)c + 1) \sum_j E_{j,t+1} B_{j,t}.$$
$$\delta_j(B_t) = E_t[M_{t,t+1}(1 - (1 - \rho)\Phi(\Delta_{t+1}(B_t)))(1 + c)E_{j,t+1}].$$

Interpretation. A currency that depreciates when firm cash flows are low reduces default risk and is valuable to borrowers even if it is unattractive to lenders.

3.2 Dominant currency condition

The dominant currency is the one with the most attractive covariance with the stock market over debt-maturity horizons, net of issuance-cost differences.

$$\text{Dominant currency} \approx \arg \max_j \text{Cov}(\text{stock market}, E_j) - \text{issuance cost}_j.$$

Interpretation. This reverses the safe-asset intuition. Borrowers prefer a currency that weakens in bad states.

3.3 Inflation-risk-premium link

$$IRP_{i,t} = \log \frac{E_t^i[P_{i,t,t+1}]}{E_t[P_{i,t,t+1}]}.$$

Under the paper’s assumptions, firms issue in dollars iff the United States has the highest inflation risk premium.

Interpretation. Monetary policy matters because countercyclical inflation can make a currency depreciate when borrowers are distressed.

3.4 Empirical regressions

$$\begin{aligned} USDshare_t &= \alpha + \beta QRP_{EUR/USD,t}^{2Y} + controls + \varepsilon_t, \\ Ret_USD_{t-h,t} &= \alpha_h + \beta_h Ret_Stock_{t-h,t} + \varepsilon_{t-h,t}, \\ 1(USD)_b &= \alpha + \beta Maturity_b + controls + FE + \varepsilon_b. \end{aligned}$$

Interpretation. The first test uses forward-looking risk premia; the second checks horizon-specific historical co-movement; the third connects the mechanism directly to maturity choice.

4 Identification and empirical design

This is a theory-guided empirical design rather than a randomized design. The strongest evidence comes from alignment across independent tests: risk premia, debt quantities, backward-looking covariances, bond maturity, yield curves, and COVID issuance.

What not to over-read. The paper does not claim investor demand, liquidity, or convenience yields are irrelevant. It shows that borrower-side default hedging can explain key patterns not captured by a purely safe-asset view.

5 Tables and figures

5.1 Figure 1: Currency denomination of international debt

Read: Dollar debt outstanding rises sharply after 2008, while the euro share falls. GBP, JPY, and CHF remain much smaller.

Interpretation. The figure establishes the dollar’s fall-and-rise pattern and motivates a dynamic theory of currency dominance.

Economic message. Dominant currency status is not purely static inertia; it changes with market conditions.

What not to over-read. This descriptive figure does not identify the mechanism.

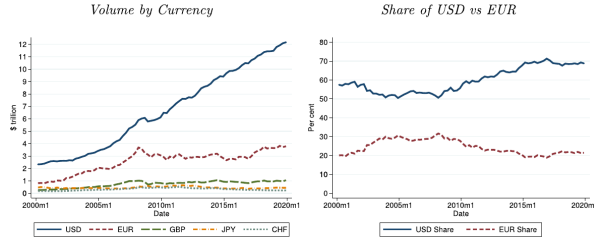


Fig. 1. Currency denomination of international debt and cross-border borrowing of non-banks (amounts outstanding). Source: Bank for International Settlements.

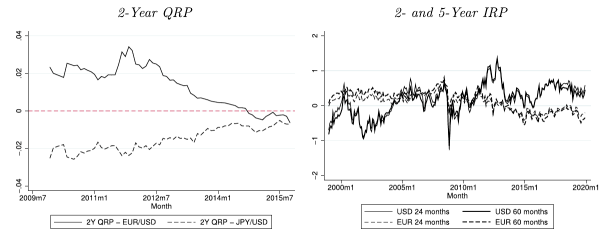


Fig. 2. Two-year quanto-implied risk premia and two- and five-year inflation risk premia in the US and eurozone. Source: QRP data are from Kremens and Martin (2019) and IRP data are from Hordahl and Tristani (2014).

5.2 Figure 2: QRP and inflation risk premia

Read: QRP falls and becomes negative; dollar IRP rises relative to euro IRP after the crisis.

Interpretation. These forward-looking prices imply that the dollar became more attractive as a borrower hedge relative to the euro.

Economic message. Market expectations about currency risk map into debt-currency choice.

What not to over-read. QRP data are short-sample and only liquid up to short maturities.

5.3 Figure 3: Gross issuance and Table 1: QRP, IRP, and currency choice

Read: Dollar issuance rises after the crisis; Table 1 shows lower QRP and lower euro IRP predict higher dollar share.

Interpretation. This links forward-looking market prices to quantities of dollar debt issuance.

Economic message. When the dollar becomes more borrower-friendly in bad states, firms issue more dollar debt.

What not to over-read. The regressions are time-series correlations over limited samples.

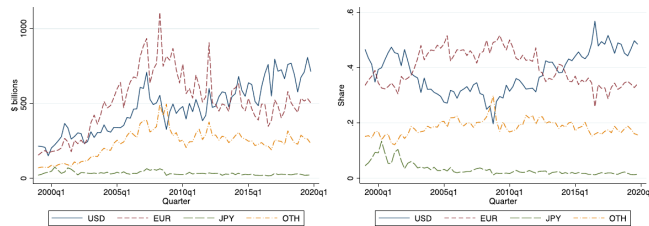


Fig. 3. Gross issuance of international debt securities. Source: BIS International Debt Securities.

Table 1
QRP, IRP, debt-currency choice.

levels, respectively. Debt-issuance data includes all sectors except the government. Latest observed values of $QRP_{\$,\2Y , $IRP_{\$, \2Y and $IRP_{\$, \2Y in a given quarter are used. $QRP_{\$, \2Y data come from Kremens and Martin (2019), and $IRP_{\$, \2Y and $IRP_{\$, \2Y are recovered using the methodology in Hordahl and Tristani (2014). Trend refers to a linear time trend and control refers to the inclusion of total issuance as a control variable.

Sample:	(1) Full USD_t^{shr}	(2) Full USD_t^{shr}	(3) Full USD_t^{shr}	(4) Full USD_t^{shr}	(5) Full USD_t^{shr}	(6) Full USD_t^{shr}
$QRP_{\$, \2Y	-3.463*** (0.314)	-1.574** (0.680)	-3.505*** (0.305)			
$IRP_{\$, \2Y				-0.00612 (0.0197)	-0.00870 (0.0254)	0.0330 (0.0232)
$IRP_{\$, \2Y				-0.201*** (0.0266)	-0.194*** (0.0406)	-0.200*** (0.0251)
Trend		X	X		X	X
Control						
Period	09q4-15q3	09q4-15q3	09q4-15q3	99q1-19q4	99q1-19q4	99q1-19q4
Observations	24	24	24	84	84	84
R-squared	0.713	0.783	0.781	0.340	0.341	0.449

expectations and therefore could occur

In this section, we rerun the regression controlling for the corporate basis to explore the global dollar share of debt issuance. The results of this regression in Table 2 are

Figure 3: Gross issuance.

Table 1: QRP, IRP, and currency choice.

5.4 Table 2: Corporate basis controls

Read: Corporate-basis controls do not eliminate the QRP and IRP evidence.

Interpretation. The results are not simply a cheaper-funding or convenience-yield story.

Economic message. Borrower hedging risk matters beyond issuance cost proxies.

What not to over-read. The corporate basis is not a complete measure of all currency-specific issuance frictions.

Table 2
QRP, IRP, corporate basis, and debt-currency choice: exchange-rate expectations versus convenience yield.
Robust standard errors are shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Debt-issuance data includes all sectors except the government. The latest observed values of $QRP_{\$/\$,t}^{2Y}$, $IRP_{\$/\$,t}^{2Y}$, and $IRP_{\$/\$,t}^{2Y}$ in a given quarter are used. $QRP_{\$/\$,t}^{2Y}$ data come from [Kremens and Martin \(2019\)](#), $IRP_{\$/\$,t}^{2Y}$ and $IRP_{\$/\$,t}^{2Y}$ are recovered using the methodology in [Hördahl and Tristani \(2014\)](#), and $Corp.Basis_t$ data come from [Liao \(2020\)](#). Different sample periods are due to differences in data availability, and multiple columns provide robustness checks by aligning sample periods.

	(1) USD_t^{dr}	(2) USD_t^{dr}	(3) USD_t^{dr}	(4) USD_t^{dr}	(5) USD_t^{dr}	(6) USD_t^{dr}	(7) USD_t^{dr}
$QRP_{\$/\$,t}^{2Y}$	-3.463*** (0.314)			-1.566** (0.718)			
$Corp.Basis_t$		-0.00113 (0.000786)	-0.00295*** (0.000450)	-0.00182** (0.000772)			-0.000846* (0.000451)
$IRP_{\$/\$,t}^{2Y}$					-0.00612 (0.0197)	0.0393 (0.0259)	0.0435* (0.0251)
$IRP_{\$/\$,t}^{2Y}$					-0.201*** (0.0266)	-0.192*** (0.0239)	-0.186*** (0.0232)
Period	09q4-15q3	03q4-16q2	09q4-15q3	09q4-15q3	99q1-19q4	03q4-16q2	03q4-16q2
Observations	24	51	24	24	84	51	51
R-squared	0.713	0.053	0.747	0.783	0.340	0.527	0.555

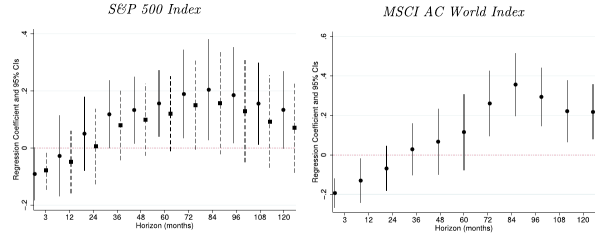


Fig. 4. The betas of the USD index returns with respect to stock market returns. The graphs report the regression coefficients β_h from the regressions [12](#). Standard errors are corrected using the Newey-West procedure with the number of lags being equal to the horizon h of returns for each respective

5.5 Figure 4: Horizon-specific dollar betas

Read: Dollar betas are negative at short horizons and positive at medium/long horizons.

Interpretation. The dollar can be a short-run safe haven and a long-run borrower hedge at the same time.

Economic message. Debt maturity is essential for understanding currency choice.

What not to over-read. Historical betas are not structural primitives.

5.6 Figure 5: VAR impulse responses and Table 3: VAR model

Read: VAR impulse responses show dollar depreciation after a negative S&P shock at medium horizons; Table 3 reports lead-lag dynamics.

Interpretation. The VAR supports the same long-horizon mechanism without relying on overlapping returns.

Economic message. The dollar's medium-run response lowers borrowers' debt burden after prolonged downturns.

What not to over-read. The VAR does not identify the fundamental structural shock behind the co-movement.

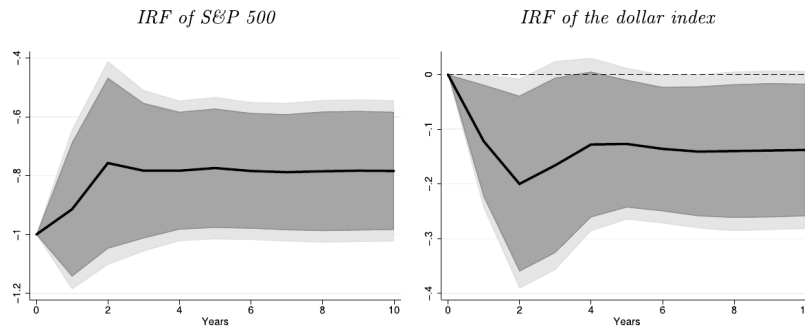


Fig. 5. Cumulative Impulse-Response Functions of a Shock to S&P 500. Figures show the cumulative impulse-response functions of a negative 1 ppt shock

Figure 5: VAR impulse responses.

Table 3
A VAR(2) model of the S&P 500 and the FRED dollar index.

Standard errors that are adjusted for small-sample degrees of freedom are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. The coefficients are from a VAR(2) model of non-overlapping annual returns on the S&P 500 and the FRED dollar index against major currencies (DTWEXM) between 1973 and 2019.

	(1) $Ret_SP500_{t-1,t}$	(2) $Ret_USD_{t-1,t}$
$Ret_SP500_{t-2,t-1}$	-0.0856 (0.147)	0.122* (0.0661)
$Ret_SP500_{t-3,t-2}$	-0.202 (0.146)	0.0494 (0.0657)
$Ret_USD_{t-2,t-1}$	0.310 (0.335)	0.322** (0.151)
$Ret_USD_{t-3,t-2}$	-0.240 (0.329)	-0.291** (0.148)
Observations	45	45
R-squared	0.0714	0.2073

Table 3: VAR model.

5.7 Table 4: Maturity and dollar issuance, Figure 6: Yield-curve slope and dollar share, and Table 5: Yield-curve slope regressions

Read: Longer-maturity bonds are more likely to be issued in dollars. Yield-curve slope differentials positively predict dollar debt shares.

Interpretation. These results directly test the maturity channel of the theory.

Economic message. Dollar debt is attractive when the maturity horizon matches the horizon over which the dollar hedges borrower default risk.

What not to over-read. Maturity can also proxy for issuer type or market segment, though fixed effects mitigate this concern.

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Table 4

Debt-maturity and debt-currency choice.

Standard errors clustered by *Country * Year* in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. $1(USD)$ is a dummy variable that takes the value of 1 if the currency of the issued bond is the dollar. $Maturity_w$ is the standardized value of maturity winsorized at 5% and 95% levels. Controls include the size of the issuance and a dummy variable for whether the bond is investment grade. *NB* refers to the sample of non-bank financials and non-financial corporations. *B* refers to the sample of banks. Columns (3), (4), (5), and (6) only include firms that issue in at least two currencies in a given month. [†] means that the sample is further restricted only to those firms that are from the US, the eurozone, Japan, Switzerland or the UK.

Sample:	(1) NB $1(USD)$	(2) B $1(USD)$	(3) NB $1(USD)$	(4) B $1(USD)$	(5) NB [†] $1(USD)$	(6) B [†] $1(USD)$
$Maturity_w$	0.0149*** (0.00339)	0.0299*** (0.00787)	0.0163 (0.0138)	0.0770*** (0.0126)	0.0350** (0.0157)	0.0734*** (0.0155)
Controls	X	X	X	X	X	X
Industry*Month FE	X	X				
Country*Month FE	X	X				
Firm*Month FE			X	X	X	X
Observations	457,266	243,657	46,562	67,213	31,857	49,268
R-squared	0.684	0.665	0.450	0.432	0.475	0.472
Mean of Dep. Var	0.556	0.534	0.376	0.348	0.430	0.350

Table 4: Maturity and dollar issuance.

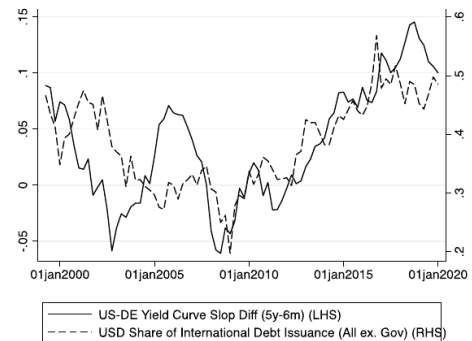


Fig. 6. Yield curve slope differentials and dollar debt share.

ditional support of the “debt view”: the fact that dollar-debt issuance is (at least partially) driven by firms’ optimal capital-structure choices.

Figure 6: Yield-curve slope and dollar share.

Table 5

US-Germany yield-curve slope differential and the share of dollar debt. Robust standard errors are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. Debt-issuance data includes all sectors except the government. $\text{Diff } f_{12,r1}^{US,DE}$ measures the difference between the US and Germany of the difference between the non-annualized interest rate over r_2 months and the non-annualized interest rate over r_1 months: $(r_2(US) - r_1(US)) - (r_2(DE) - r_1(DE))$. The sample period is 1999Q1–2019Q4. Each column shows the coefficient on $\text{Diff } f_{12,r1}^{US,DE}$ of the regression of USD^{hr} on $\text{Diff } f_{12,r1}^{US,DE}$, for different values of r_1, r_2 , and controls. Controls include a time trend and total international debt issuance excluding issuance by governments in a given quarter.

Dep. Var:	(1) USD^{hr}	(2) USD^{hr}	(3) USD^{hr}	(4) USD^{hr}	(5) USD^{hr}	(6) USD^{hr}	(7) USD^{hr}
Indep. Var:	$\text{Diff } f_{12,6,1}^{US,DE}$	$\text{Diff } f_{24,6,1}^{US,DE}$	$\text{Diff } f_{60,6,1}^{US,DE}$	$\text{Diff } f_{120,6,1}^{US,DE}$	$\text{Diff } f_{24,12,1}^{US,DE}$	$\text{Diff } f_{60,12,1}^{US,DE}$	$\text{Diff } f_{120,12,1}^{US,DE}$
	2.539*** (0.873)	1.016*** (0.297)	0.515*** (0.119)	0.279*** (0.0637)	1.633*** (0.447)	0.612*** (0.134)	0.303*** (0.0673)
Controls	X	X	X	X	X	X	X
Observations	84	84	84	84	84	84	84
R-squared	0.618	0.629	0.656	0.664	0.635	0.661	0.666

Dollar against major currencies

Inflation risk premia

Table 5: Yield-curve slope regressions.

5.8 Figure 7: COVID dollar and IRP dynamics and Table 6: Post-COVID issuance

Read: During COVID, the dollar depreciates against major currencies and dollar IRP rises relative to euro IRP. Post-COVID bonds are more likely to be dollar denominated.

Interpretation. The COVID episode shows that changes in risk expectations can shift dollar issuance quickly.

Economic message. Dominant currency status can respond to expectations, not only slow-moving network effects.

What not to over-read. COVID also involved extraordinary policy and liquidity interventions, so it is not a clean natural experiment.

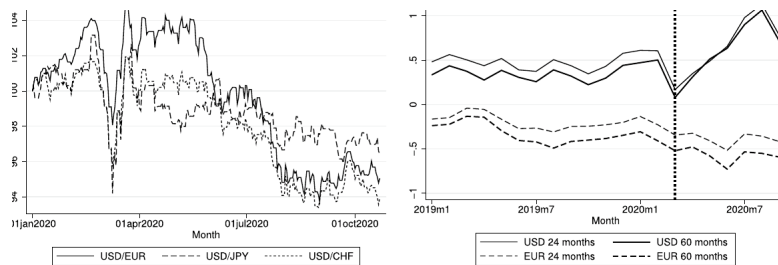


Fig. 7. The dollar and inflation risk premia during the COVID-19 crisis. Sources: Datastream, Hürdahl and Tristani (2014), authors' calculations.

the Swiss franc. Unlike previous episodes of global shocks, the dollar depreciated against these currencies without a

the dollar's international role in denominating debt contracts, consistent with Proposition 3.

Figure 7: COVID dollar and IRP dynamics.

Table 6

Debt-issuance and debt-currency choice: Pre- and post-COVID period. Robust standard errors are in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. The sample in columns (1), (2), and (3) is restricted to issuance in five currencies, USD, EUR, JPY, CHF, and GBP, in 2019 and 2020 (data retrieved in October 2020 from Dealogic). $\mathbb{1}(\text{USD})$ is a dummy variable that takes the value of 1 if the currency of the issued bond is the dollar. $\mathbb{1}(\text{Post_COVID})$ is a dummy that is 1 if the issue date is after April 1, 2020. Controls include the winorized maturity at 5%, the size of the issuance, and a dummy variable for the investment-grade status of the bond. All columns include a firm fixed effect. B refers to the sample of banks, $NBFI$ refers to the sample of non-bank financials, and NFC refers to the sample of non-financial corporations. Column (4) includes all these sectors. The $\uparrow$ in column (4) means observations are limited to dollar and euro issuance by firms outside the US and the euro area.

Sample:	(1) B $\mathbb{1}(\text{USD})$	(2) NBFI $\mathbb{1}(\text{USD})$	(3) NFC $\mathbb{1}(\text{USD})$	(4) Full $\uparrow$ $\mathbb{1}(\text{USD})$
$\mathbb{1}(\text{Post_COVID})$	0.0269** (0.0122)	0.00772** (0.00375)	0.0250*** (0.00841)	0.0325*** (0.0121)
Controls	X	X	X	X
Firm FE	X	X	X	X
Observations	4978	24,506	5866	6472
R-squared	0.751	0.924	0.875	0.666

Table 6: Post-COVID issuance.

6 Short summary

- The paper proposes a borrower-driven debt view of the dollar.
- The dominant currency is the one that hedges borrower default risk at debt-maturity horizons.

- Forward-looking QRP and IRP measures support the dollar’s post-crisis dominance.
- Historical evidence shows the dollar is safe at short horizons but risky at longer horizons.
- Longer-maturity bonds are more likely to be dollar denominated.
- COVID evidence is consistent with a quick increase in dollar issuance when risk expectations changed.

7 Conclusion

7.1 Core conclusion

Eren and Malamud argue that the dollar dominates international debt markets because it is useful to borrowers. It weakens over horizons relevant for debt repayment when global conditions deteriorate, reducing default risk.

7.2 Economic intuition

A currency that appreciates in bad states is good for bondholders but bad for borrowers. A currency that depreciates in bad states lowers the real repayment burden and supports higher debt capacity.

7.3 Evidence synthesis

The paper combines evidence from QRP, IRP, debt shares, historical dollar-stock betas, VAR impulse responses, maturity regressions, yield-curve regressions, and COVID issuance patterns.

7.4 Policy implications

Dominant-currency status may depend on monetary policy and inflation cyclicalities. A currency need not be the safest haven to become dominant in debt issuance; it may need to be risky in a way that helps borrowers.

7.5 Limitations and future research

The paper abstracts from safe-asset demand, dealer balance sheets, trade invoicing, bank regulation, and firm-level operating exposures. Future work can integrate the debt view with trade, banking, reserves, and financial intermediation.

7.6 Takeaway

Debt currency is a capital-structure choice. The dollar’s dominance is partly a result of its long-horizon risk properties, not only liquidity and inertia.